Poster 248: Lateral extra-articular tenodesis augmentation of anterior cruciate ligament reconstruction does not impact cartilage health in the patellofemoral joint based on post-operative two year qualitative and quantitative MRI analysis

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Objectives: Lateral extra-articular tenodesis (LET) augmentation of anterior cruciate ligament reconstruction (ACLR) has been shown in multiple clinical and biomechanics studies to be an effective procedure to control rotational laxity, reducing failure rates. However, it is uncertain whether augmentation with an iliotibial band based LET may cause increased contact pressure in the patellofemoral joint (PFJ), resulting in cartilage damage. The purpose of the current study was to assess PFJ cartilage status in patients that underwent ACLR with or without LET augmentation using magnetic resonance imaging (MRI). It was hypothesized that 1) the addition of LET at the time of ACLR would have no effect on cartilage health on post-operative two-year MRI; 2) higher cartilage relaxation values would be associated with worse patient reported and functional outcomes at two years post-operatively.

Methods: A subset of patients from the STABILITY 1 randomized controlled trial were included. All patients underwent primary ACLR with a hamstring autograft. The patients were randomized either to receive LET augmentation (modified Lemaire technique) or not. Cartilage status in the PFJ between ACLR alone group and ACLR+LET group were compared by quantitative MRI (qMRI) analysis of T2 and T1 rho sequences (figure 1), and qualitative assessment by ACL osteoarthritis score (ACLOAS) on post-operative two year MRI for surgical and contralateral non-surgical knees. Objective functional outcomes (range of motion (ROM), hop test and isokinetic strength) and patient reported outcome measures including Knee injury Osteoarthritis and Outcome Score (KOOS) and International Knee Documentation Committee (IKDC) score were attained. Descriptive statistics were calculated for each group using means and standard deviations (SD) for continuous variables and proportions for categorical variables. ANCOVA was used to determine the adjusted mean difference in T1rho and T2 relaxation times between groups for each compartment in the patella and trochlea. Spearman's rho was used to assess the correlation between relaxation times within each compartment and patient-reported and functional outcomes at two-years postoperative. Statistical significance was set at P = 0.05.

Results: Ninety-two patients (43 patients in ACLR group, mean age (±SD) 18.9±3.2 years, 60.5% female, and 49 patients in ACLR+LET group, 18.7±3.2 years, 63.3% females) were included. No significant differences were seen in patient background, meniscal treatment and cartilage status at the time of surgery (table 1). No significant differences were seen in the mean values (milliseconds) for qMRI for adjusted T1 rho relaxation times in the medial patella (MP) (47.8 vs 47.3), central patella (CP) (45.5 vs 44.1), and lateral patella (LP) (48.2 vs 47.3), and medial trochlea (MT) (54.7 vs 56.4), central trochlea (CT) (53.3 vs 53.1), and lateral trochlea (LT) (54.9 vs 53.9), and adjusted T2 relaxation times in MP (42.2 vs 43.2), CP (42.5 vs 42.7), LP (43.5 vs 43.0), MT (50.9 vs 50.9), CT (51.1 vs 52.0), LT (52.1 vs 52.6) between ACLR and ACLR+LET groups. Similarly, no difference in ACLOAS scores (overall score 10 vs 11, p = 0.99) were observed. An increase in medial patellar T2 relaxation times were associated with decreasing IKDC score (p = 0.046), KOOS

symptoms subscale (p = 0.01), and total KOOS score (p = 0.01). Increasing limb symmetry on the hop test was associated with increasing T1ρ (p = 0.04) and T2 (p = 0.01) relaxation in the lateral trochlea.

Conclusions: No difference was found in patellofemoral cartilage health between knees undergoing primary ACLR with hamstring tendon autograft with or without LET at two years post-operative. Statistically significant correlations were found between qMRI relaxation times and functional outcome scores and PROMs, however the correlations were weak and the clinical significance of these changes are undetermined.

Characteristic	ACLR only (n = 43)	ACLR + LET (n = 49)	P-value
Age	18.9 ± 3.2	18.7 ± 3.2	0.79
Female sex, n (%)	26 (60.5)	31 (63.3)	0.84
Height (inches), mean ± SD	67.9 ± 3.7	67.1 ± 3.5	0.27
Weight (pounds), mean ± SD	155.5 ± 33.2	158.9 ± 33.2	0.63
BMI (kg/m^2), mean ± SD	23.5 ± 3.4	24.6 ± 3.5	0.13
Non-contact injury, n (%)	26 (60.5)	26 (53.1)	0.49
Tear chronicity (months), median (IQR)	4 (4)	5 (11)	0.27
Concomitant meniscal repair			0.31
Medial meniscus repair	14 (32.6)	12 (24.5)	
Lateral meniscus repair	7 (16.3)	4 (8.2)	
Both menisci repaired	6 (14.0)	5 (10.2)	
Concomitant meniscal excision			0.05
Medial meniscus excision	1 (2.3)	1 (2.0)	
Lateral meniscus excision	8 (18.6)	2 (4.1)	
Both menisci excised	0 (0)	2 (4.1)	
Cartilage lesions (ICRS grade 3+), n (%)	3 (7.0)	2 (4.1)	0.66
Patellar or trochlear lesions	0 (0)	1 (2.2)	
Returned to sport post-operative, n (%)	40 (93.0)	44 (89.8)	0.42
Postoperative sport level and demand, n (%)			0.82
Low risk, low level	5 (11.6)	7 (14.3)	
Low risk, high level	6 (13.6)	4 (8.2)	
High risk, low level	8 (18.6)	12 (24.5)	
High risk, high level	20 (46.5)	21 (42.9)	
RTS exposure time (months), mean ± SD	12.9 ± 5.3	11.1 ± 5.1	0.11

Table 1. Patient background and surgical interventions of ACLR only and ACLR + LET groups. ACLR (anterior cruciate ligament reconstruction), LET (lateral extra-articular tenodesis), BMI (body mass index), IQR (interquartile range), SD (standard deviation), RTS (return to sport), ICRS (international cartilage repair society). Statistical significance set as p < 0.05.

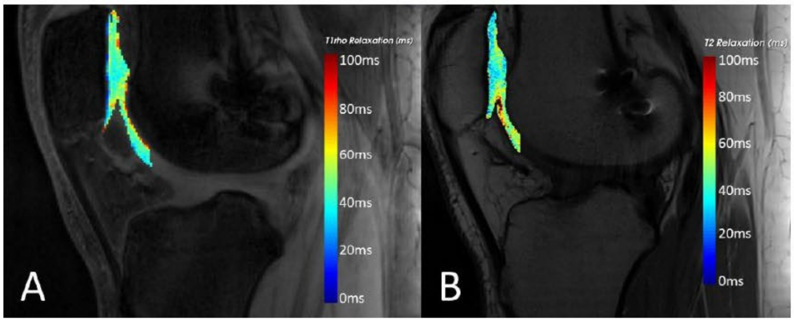

Figure 1. Color-coded sagittal T1ρ (A) and T2 (B) images for relaxation time of the patella and trochlea

The Orthopaedic Journal of Sports Medicine, 11(7)(suppl 3)
DOI: 10.1177/2325967123S00228